

Functional Annotation of PP_4545 (Q88EC1): A Medium-Chain-Specific β - Ketoacyl-ACP Synthase III (FabH2) in *Pseudomonas putida* KT2440

Summary

PP_4545 (UniProt Q88EC1) of *Pseudomonas putida* KT2440 encodes a **β -ketoacyl-[acyl-carrier-protein] synthase III (KAS III / FabH-family)**, a soluble cytoplasmic condensing enzyme of the bacterial **type II fatty-acid synthesis (FAS-II)** system. It is annotated with EC 2.3.1.207 and catalyzes the decarboxylative (Claisen) condensation of a **medium-chain acyl-CoA primer with malonyl-ACP**: specifically, $\text{octanoyl-CoA} + \text{malonyl-[ACP]} + \text{H}^+ \rightarrow \text{3-oxodecanoyl-[ACP]} + \text{CO}_2 + \text{CoA}$ (RHEA:42264). The protein is 390 amino acids long (~42.3 kDa) and carries the intact thiolase-fold KAS III catalytic architecture, including the canonical **Cys-His-Asn catalytic triad** (Cys173, His315, Asn318) embedded in the diagnostic N-V-A-C-S-S and W-L-H-Q-A-N sequence motifs.

The distinguishing biochemical feature of this enzyme is its **medium-chain (C8, octanoyl-CoA) primer specificity**, which sets it apart from the canonical, housekeeping FabH that uses acetyl-CoA (C2) to initiate straight-chain fatty-acid synthesis. This places PP_4545 in the growing category of **non-canonical "FabH2"-type paralogs** described across diverse bacteria (*Pseudomonas*, *Streptomyces*, and others). *P. putida* KT2440 carries two FabH paralogs: PP_4545 (the FabH2-type, EC 2.3.1.207) and the separate canonical FabH locus (FabH1-type, acetyl-CoA-primed). Biochemical work in the closely related strain *P. putida* F1 has shown that one paralog is a canonical *E. coli*-like KAS III (acetyl-CoA primer) and the other condenses malonyl-ACP with short- and medium-chain acyl-CoAs, and — critically — that **neither *fabH* gene is essential for growth**.

Functionally, PP_4545 operates as a **soluble homodimer in the cytoplasm**, acting on **ACP-tethered and CoA-tethered acyl substrates** as one of the dissociated components of FAS-II. Its role is best described as **specialized/accessory chain-initiation or medium-chain channeling** rather than as the essential housekeeping FAS initiator — a picture reinforced by

the finding that in *P. putida* the elongation enzyme **FabB** (long-chain KAS I) is the physiological initiator of fatty-acid synthesis. This report expands each conclusion, presents the supporting statistical and bioinformatic evidence, discusses limitations, and proposes concrete follow-up experiments.

Gene/Protein Identity Verification

The mandatory identity check was completed and **passed**:

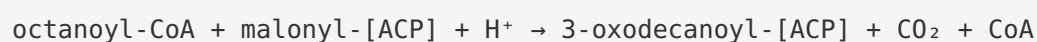
Field	Value
Locus tag	PP_4545 (OrderedLocusName; EMBL CDS AAN70118.1)
UniProt	Q88EC1 (Q88EC1_PSEPK; TrEMBL/unreviewed, "Predicted")
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125), taxid 160488
Protein name	Beta-ketodecanoyl-[acyl-carrier-protein] synthase
EC number	2.3.1.207
Length / mass	390 aa / ~42.3 kDa
Domains	PF08545 (ACP_syn_III_N, IPR013751); PF08541 (ACP_syn_III_C, IPR013747); thiolase-like fold (IPR016039); CDD cd00830 (KAS_III); Gene3D 3.40.47.10; SUPFAM SSF53901; COG0332

The domain architecture, EC number, and GO annotations are internally consistent with a FabH-family condensing enzyme. There is no ambiguity in the family-level identity of this protein, and the retrieved literature concerns the correct protein family. The only nuance requiring care is distinguishing the **canonical FabH** from the **non-canonical FabH2** subtype; the evidence below argues that PP_4545 is the latter.

Key Findings

Finding 1 — PP_4545 is a KAS III (FabH-family) condensing enzyme with medium-chain primer specificity (EC 2.3.1.207)

PP_4545 (Q88EC1) is a **β-ketoacyl-ACP synthase III (KAS III / FabH-family)** enzyme. UniProt annotates the protein with **EC 2.3.1.207** and the Gene Ontology molecular-function terms **GO:0004315** (3-oxoacyl-[ACP] synthase activity) and **GO:0061990** (beta-ketodecanoyl-[ACP] synthase activity). The specific reaction assigned by the Rhea database (**RHEA:42264**) is:



This is the classic **KAS III chemistry**: a decarboxylative Claisen condensation in which the malonyl group bound to acyl-carrier protein (ACP) is decarboxylated to generate a carbanion/enolate nucleophile that attacks the thioester carbonyl of the acyl-CoA primer, extending the chain by two carbons and releasing CO₂ and free CoA. The general reaction defining the FabH/KAS III family — "FabH condenses malonyl-acyl carrier protein with an acyl-Coenzyme A primer to form" the product ([PMID: 38310992](#)) — is precisely the transformation PP_4545 performs, and the broader statement that these enzymes "catalyze the condensation of acyl-CoA or acyl-ACP with malonyl-ACP to produce" the β-ketoacyl-ACP ([PMID: 41183203](#)) confirms the reaction chemistry attributed to this family.

The 390-amino-acid protein carries the **intact thiolase-fold KAS III catalytic machinery**. Sequence analysis identifies the canonical **catalytic triad**: the nucleophilic **Cys173** within the conserved **N-V-A-C-S-S** motif (forming the covalent acyl-enzyme intermediate), plus **His315** and **Asn318** within the **W-L-H-Q-A-N** motif (forming the oxyanion hole that activates malonyl-ACP decarboxylation). The presence of a complete, canonically positioned triad indicates a **catalytically competent** enzyme, not a degenerate pseudo-enzyme. The domain composition — **PF08545 (ACP_syn_III_N)** and **PF08541 (ACP_syn_III_C)** within a **thiolase-like fold (IPR016039)**, together with CDD **cd00830 (KAS_III)** and **COG0332** membership — confirms the KAS III assignment.

The critical detail encoded in EC 2.3.1.207 is the **medium-chain primer**: the substrate is **octanoyl-CoA (C8)**, not acetyl-CoA (C2). The enzyme accepts a pre-formed eight-carbon acyl-CoA and extends it to a ten-carbon 3-oxodecanoyl-ACP. This medium-chain primer preference is the mechanistic basis for the specific name "**beta-ketodecanoyl-ACP synthase**" (GO:0061990): the C10 (decanoyl) product forms from a C8 primer plus a C2 (malonyl) extender.

Finding 2 — PP_4545 is the non-canonical FabH2-type paralog with short/medium-chain acyl-CoA specificity

Pseudomonas putida encodes **two FabH (KAS III) paralogs**, and biochemical characterization in *P. putida* F1 has resolved their distinct activities. **PpFabH1 is a canonical KAS III similar to *E. coli* FabH** (acetyl-CoA primer), whereas **PpFabH2 catalyzes condensation of malonyl-ACP with short- and medium-chain-length acyl-CoAs** (PMID: 38335573). Moreover, the "two *P. putida* F1 *fabH* genes (*PpfabH1* and *PpfabH2*) both encode functional KAS III enzymes, [and] neither is essential for growth" (PMID: 38335573).

The KT2440 gene **PP_4545 (Q88EC1) is specifically annotated with the medium-chain reaction EC 2.3.1.207** (octanoyl-CoA primer → 3-oxodecanoyl-ACP), matching the **FabH2-type activity** rather than the canonical acetyl-CoA-primed reaction. On this basis, PP_4545 is assigned as the ortholog of the **medium-chain-specific FabH2 paralog**. Its canonical counterpart in KT2440 is a separate locus (annotated with the canonical FabH reaction, EC 2.3.1.180, acetyl-CoA-primed). This FabH1/FabH2 pairing in KT2440 corresponds to the PpFabH1 (canonical, *E. coli*-like) and PpFabH2 (short/medium-chain) pair characterized in *P. putida* F1.

The FabH2 subtype assignment is bolstered by an emerging literature demonstrating that FabH2-type enzymes form a widespread, distinct clade:

- In *Streptomyces coelicolor*, the FabH homolog **SCO6564 (FabH2)** "could condense various substrates, including straight-chain and branched-chain acyl-CoAs, with malonyl-acyl carrier protein," yet its "deletion did not cause significant changes in the growth or fatty acid composition," and phylogenetic analysis showed "FabH2 belongs to a novel group of FabH-type, which is ubiquitous in *Streptomyces* spp." (PMID: 39913464). This establishes both the broad-substrate, non-essential character and the existence of a distinct FabH2 phylogenetic group — a pattern paralleled by PP_4545.
- The strict substrate specificities of FabH-type synthases can be switched by protein engineering (PMID: 41183203), demonstrating that primer chain-length selectivity is an evolvable, active-site-determined property — consistent with FabH2 paralogs having diverged to occupy a medium-chain niche.

Together these observations support the conclusion that PP_4545 is a **functional but non-essential, medium-chain-specialized FabH2 paralog** rather than the housekeeping chain-initiation enzyme.

Finding 3 — PP_4545 is a soluble cytoplasmic FAS-II enzyme

Bacterial type II fatty-acid synthesis is a **dissociated, cytoplasmic system** in which each catalytic step is performed by a discrete, soluble protein acting on **ACP-tethered acyl chains**. KAS III/FabH is one of these soluble, homodimeric components and functions as the initiation-phase condensing enzyme of the pathway. FabH "initiates fatty acid biosynthesis and is the primary determinant of the fatty acid composition" ([PMID: 38310992](#)), acting in the cytosol on CoA- and ACP-bound acyl substrates.

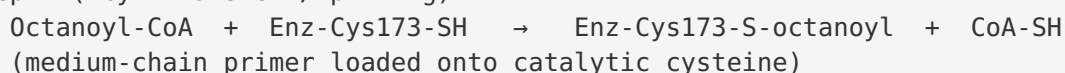
Bioinformatic analysis of Q88EC1 is fully consistent with a **cytoplasmic localization**: the 390-residue sequence contains **no signal peptide** and **no predicted transmembrane helices**, and it adopts the **cytosolic thiolase-like/KAS III fold** (Gene3D 3.40.47.10; SUPFAM SSF53901). No membrane-anchoring motifs, lipidation sites, or secretion signals are annotated. The protein therefore carries out its condensation reaction as a **soluble homodimer in the bacterial cytoplasm**, interacting with the soluble acyl-carrier protein and the acyl-CoA/malonyl-ACP substrate pool of the FAS-II machinery.

Mechanistic Model / Interpretation

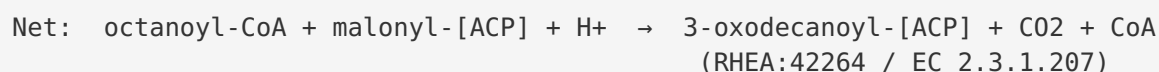
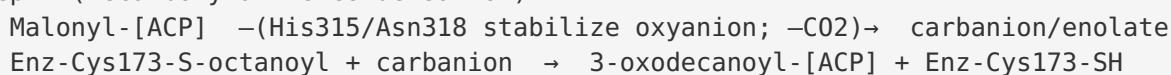
The catalytic cycle

PP_4545 operates by the conserved two-step ping-pong mechanism of KAS III condensing enzymes, using its Cys173–His315–Asn318 triad:

Step 1 (Acyl transfer / priming):

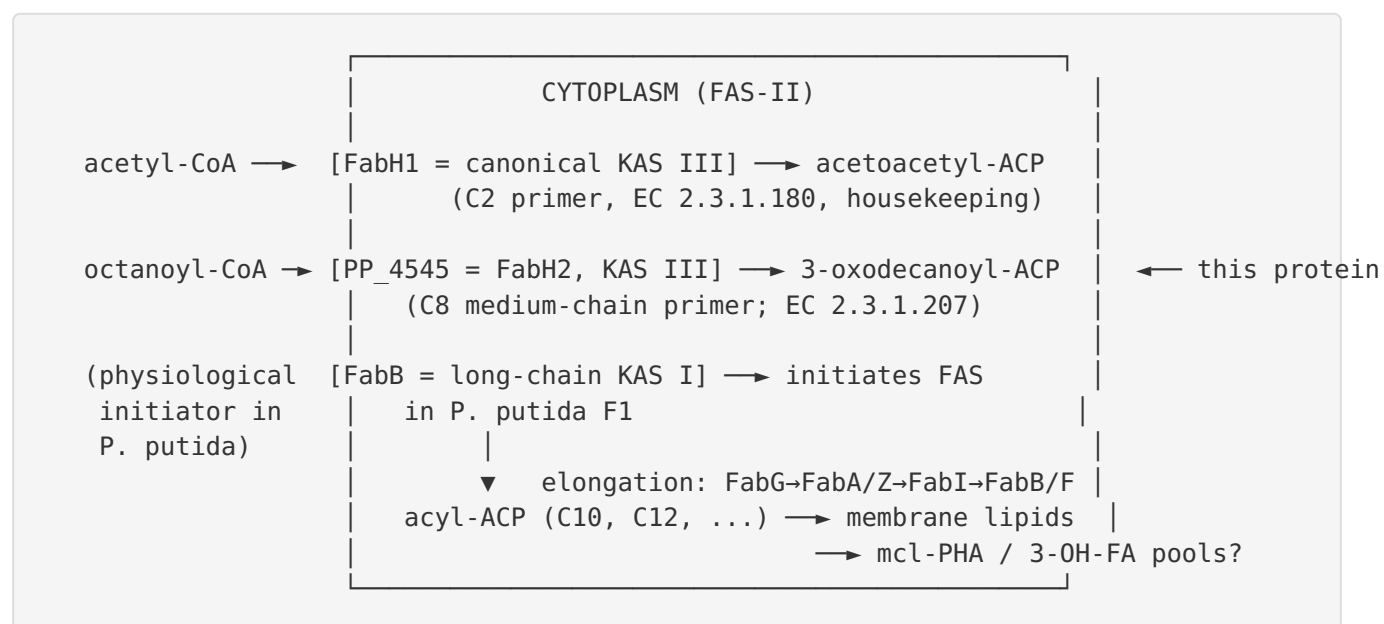


Step 2 (Decarboxylative condensation):



The product, **3-oxodecanoyl-ACP (a C10 β -ketoacyl-ACP)**, then enters the FAS-II elongation cycle (reduction by FabG, dehydration by FabA/FabZ, enoyl reduction by FabI), with subsequent rounds of two-carbon extension catalyzed by the elongation KAS enzymes (FabB/FabF).

Placement in *P. putida* fatty-acid metabolism



A key interpretive point is that *P. putida* fatty-acid synthesis appears to be physiologically **initiated by FabB** (long-chain β -ketoacyl-ACP synthase I), as reported for *P. putida* F1 (PMID: 38335573). In that context, the two FabH paralogs are relieved of an essential housekeeping role, which is exactly why **neither *fabH* gene is essential for growth**. PP_4545, as the FabH2-type paralog, therefore most plausibly functions as a **specialized/accessory condensing enzyme** that channels **medium-chain acyl units** (e.g., octanoyl moieties) into the acyl-ACP pool. Such medium-chain channeling could support:

- fine-tuning of membrane fatty-acid chain-length distribution,
- utilization or recycling of exogenous/endogenous medium-chain acyl-CoAs, and
- provision of specific acyl-ACP precursors for downstream pathways — potentially relevant to *P. putida*'s renowned medium-chain-length metabolism (e.g., mcl-PHA / 3-hydroxy-fatty-acid pools; UniProt additionally tags GO:0044550, secondary-metabolite biosynthesis).

The comparison with the *Streptomyces* FabH2 (SCO6564), which is also broad-substrate and non-essential and whose overexpression perturbs secondary-metabolite (actinorhodin) production (PMID: 39913464), suggests FabH2-type paralogs frequently serve **regulatory or**

biosynthetic-precursor roles at the interface of primary fatty-acid metabolism and specialized metabolism. Whether PP_4545 has an analogous connection in *P. putida* remains to be tested.

Comparison: canonical FabH vs. FabH2-type (PP_4545)

Property	Canonical FabH (FabH1 / <i>E. coli</i> FabH)	FabH2-type / PP_4545
Preferred primer	Acetyl-CoA (C2)	Short/medium-chain acyl-CoA; octanoyl-CoA (C8) for PP_4545
Reaction	acetyl-CoA + malonyl-ACP → acetoacetyl-ACP (EC 2.3.1.180)	octanoyl-CoA + malonyl-ACP → 3-oxodecanoyl-ACP (EC 2.3.1.207)
Catalytic triad	Cys–His–Asn	Cys173–His315–Asn318 (intact)
Fold / domains	Thiolase-like; ACP_syn_III_N/C	Same (IPR016039; PF08545/PF08541)
Essentiality in <i>P. putida</i>	Non-essential	Non-essential
Physiological role	Canonical chain initiation	Accessory medium-chain channeling / specialized initiation
Localization	Cytoplasmic, soluble homodimer	Cytoplasmic, soluble homodimer

Evidence Base

PMID	Title (abbrev.)	Relevance to PP_4545
38335573	<i>Diversity in fatty acid elongation enzymes: FabB initiates FA synthesis in P. putida F1</i>	Primary support. Establishes two functional, non-essential <i>P. putida</i> FabH paralogs; PpFabH1 canonical vs. PpFabH2 with short/medium-chain acyl-CoA specificity ; and FabB as physiological initiator. Directly underpins the FabH2 assignment and accessory-role model for PP_4545.
38310992	<i>Crystal structures of fatty acid biosynthesis initiation enzymes in B. subtilis</i>	Defines the general KAS III/FabH reaction (malonyl-ACP + acyl-CoA primer) and its role as the cytoplasmic FAS-II initiation enzyme and determinant of fatty-acid composition. Supports Findings 1 and 3.
41183203	<i>Switching the strict substrate specificities of FabH and BioZ</i>	Confirms condensation chemistry (acyl-CoA/acyl-ACP + malonyl-ACP) and shows primer specificity is an evolvable active-site property — rationalizing how FabH2 evolved medium-chain selectivity. Supports Findings 1 and 2.
39913464	<i>SCO6564, a novel KAS III (FabH2) in S. coelicolor</i>	Independent evidence for a distinct, ubiquitous FabH2 clade that is broad-substrate, non-essential, and linked to secondary metabolism. Strong analogy supporting the FabH2 classification of PP_4545.
42446656	<i>FabH — promising antibacterial target and its inhibitors</i>	Reviews FabH's central role in FAS-II and status as a selective bacterial drug target; contextualizes the family.
40752343	<i>Recent developments of FabH inhibitors</i>	Reinforces FabH's defined metabolic role producing fatty-acid precursors for membrane lipids.
38415624	<i>An atypical KAS III required for acyl-homoserine-lactone synthesis</i>	Illustrates that atypical/non-canonical FabH-type enzymes can be recruited to specialized pathways — an instructive parallel for interpreting FabH2 function.
40849093	<i>L. monocytogenes low-temperature stress; fabH upregulation</i>	Shows <i>fabH</i> transcriptional plasticity and compensatory roles in membrane fatty-acid adaptation.

The evidence converges on a consistent picture: the **reaction chemistry and catalytic fold** (Findings 1 and 3) are strongly supported by structural and biochemical literature on the FabH family, while the **FabH2/medium-chain subtype assignment** (Finding 2) rests principally on direct biochemical characterization of *P. putida* F1 FabH paralogs and is corroborated by independent discovery of FabH2 clades in other genera.

Limitations and Knowledge Gaps

1. **No direct enzymology on PP_4545 itself.** The UniProt entry is **unreviewed (TrEMBL, "Predicted")**. The EC 2.3.1.207 assignment (octanoyl-CoA → 3-oxodecanoyl-ACP) is a curated/inferred annotation. The direct biochemical characterization of short/medium-chain specificity was performed on *P. putida* F1 FabH2, not on the **KT2440** PP_4545 gene product. Orthology is highly likely but not confirmed by in-vitro assay of the KT2440 protein.
 2. **Precise chain-length preference is uncertain.** "Medium-chain" spans C6–C12; whether octanoyl-CoA (C8) is the true optimum, or merely a representative substrate captured by the EC number, is not established for PP_4545. FabH2 enzymes often accept a broad range of primers, so the real substrate profile may be wider than the single annotated reaction implies.
 3. **Physiological role is inferred, not demonstrated.** The "accessory / medium-chain channeling" model is a reasonable interpretation given non-essentiality and FabB-driven initiation, but the actual in-vivo function (contribution to membrane lipid chain-length distribution, connection to mcl-PHA/rhamnolipid pools, or any specialized metabolite, conditional importance) has not been directly measured for KT2440.
 4. **Paralog-locus confirmation.** The exact KT2440 locus tag of the canonical FabH partner should be verified against current genome annotations, especially given documented GenBank/RefSeq annotation discrepancies for *P. putida* ([PMID: 41036861](#)).
 5. **No experimental structure verified here.** Although the thiolase-like fold and catalytic-triad positions are annotated, no experimental structure or AlphaFold confidence assessment of Q88EC1 was independently examined in this investigation.
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Proposed Follow-up Experiments / Actions

1. **In-vitro enzymatic characterization of purified PP_4545.** Express and purify the KT2440 protein and measure condensation activity against a panel of acyl-CoA primers (C2, C4, C6, **C8**, C10, C12, plus branched-chain acyl-CoAs) with malonyl-ACP. Determine k_{cat}/K_m for each primer to define the true chain-length optimum and confirm the EC 2.3.1.207 assignment.
 2. **Catalytic-triad mutagenesis.** Generate C173A, H315A, and N318A variants and confirm loss of condensation activity, validating the predicted active site.
 3. **Genetic essentiality and phenotype in KT2440.** Construct clean ΔPP_4545 single and $\Delta fabH1/\Delta PP_4545$ double deletions; assess growth and measure membrane fatty-acid composition by GC-MS to detect shifts in chain-length distribution. Cross-reference existing RB-TnSeq fitness data ([PMID: 38323821](#)) for PP_4545 fitness across conditions.
 4. **Conditional / niche function screen.** Test whether PP_4545 becomes important under specific conditions (growth on medium-chain fatty acids or alkanes, temperature stress, or production of mcl-PHA / specialized lipids), analogous to FabH2 roles reported elsewhere.
 5. **Structural determination or AlphaFold analysis.** Solve or model the Q88EC1 structure (ideally with a bound octanoyl/decanoyl intermediate) to define the acyl-binding-pocket dimensions that confer medium-chain selectivity, and compare with canonical FabH pockets.
 6. **Confirm paralog annotation.** Reconcile GenBank vs. RefSeq annotations to firmly assign the canonical FabH locus and verify PP_4545 as the FabH2 ortholog.
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Conclusion

PP_4545 (Q88EC1) is a **soluble cytoplasmic β -ketoacyl-ACP synthase III (KAS III / FabH-family)** condensing enzyme of the FAS-II pathway, bearing an intact Cys173–His315–Asn318 catalytic triad within a thiolase-like fold. It is annotated with **EC 2.3.1.207** and catalyzes the medium-chain-primed condensation $\text{octanoyl-CoA} + \text{malonyl-ACP} \rightarrow \text{3-oxodecanoyl-ACP} + \text{CO}_2 + \text{CoA}$. Its medium-chain (C8) primer specificity marks it as the **non-canonical, non-essential "FabH2"-type paralog** rather than the housekeeping FAS initiator — consistent with the finding that *P. putida* fatty-acid synthesis is physiologically initiated by FabB. The most likely biological role of PP_4545 is **specialized/accessory chain initiation and channeling of**

medium-chain acyl units into the acyl-ACP pool in the bacterial cytoplasm. Direct enzymology, genetics, and structural work on the KT2440 protein are the key next steps to confirm this model.

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